PAPER_1623_SYNODIC_MONTH_29_53

Daniel T. Murphy

PAPER_1623 — Lunar Synodic Month = T_synod = D_crit + D − F·D − F·Φ + F²·K = 29.5375 days

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Astronomy
Date: June 18, 2026
Status: CLOSED — 0.025% closure

Observation

PAPER_1209Z_S582 (Astronomy Unified Proof Set) gives the formula:

```
T_synod = D_crit + D − F·D − F·Φ + F²·K = 29.5375 days
```

Residual: **0.025%**.

UQFF Closed Identity

```
T_synod = D_crit + D − F·D − F·Φ + F²·K = 29.5375 days 0.025%
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209Z_S582
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "synodic_month_29_53"})`
